

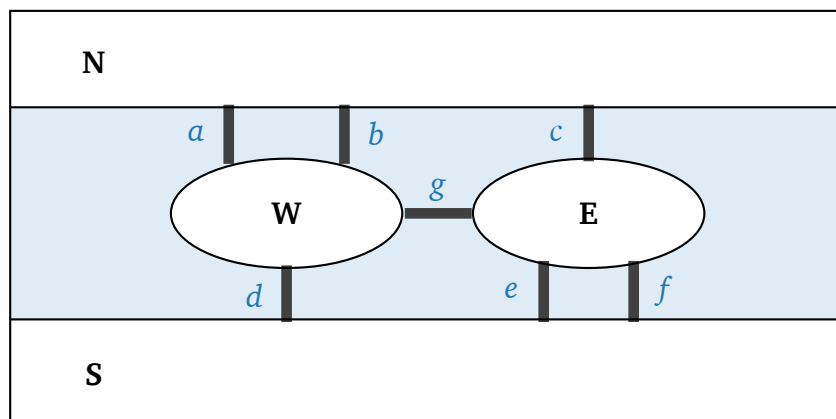
Cross Every Bridge Once, Then Teach a Computer to Do It

Work through this **before** the lecture. You are not supposed to know the technical words yet. Every word you need is defined on the sheet itself, and always *after* you have already built the thing it names. Use a pencil.

Part 1 — Seven bridges

A university campus sits on both banks of a creek. There are two islands in the middle. Seven footbridges, *a* to *g*, join the four pieces of land: the north bank **N**, the south bank **S**, the west island **W**, and the east island **E**.

You are a campus tour guide. Every footbridge has a donor's name on it, so your tour must cross **every bridge exactly once**.



Question 1(a): Find a route that crosses all seven bridges, each one exactly once. You may start on any piece of land and finish on any piece of land. You may step on the same piece of land more than once.

Write your route like this: land, bridge, land, bridge, ... The first two steps are done for you. Start on N, cross bridge *a* to reach W, then cross bridge *b* back to N.

N *a* W *b* N _____

Question 1(b): Draw an **eighth** bridge on the map above. Call it *h*, and draw it between W and E, just below bridge *g*. The map now has eight bridges. Call it **Map B**.

Try again on Map B: cross all eight bridges, each one exactly once. Write your best attempt, and write which bridge was left over at the end.

Question 2: Fill in the table below for the **seven-bridge** map, using the route you found in Question 1(a).

For each piece of land, count three things:

- **arrivals** — how many times you walked *onto* it,
- **departures** — how many times you walked *off* it,
- **bridges** — how many bridges touch it on the map.

Worked example. Take the route N *a* W *b* N *c* E *g* W *d* S *e* E *f* S. At N you leave by *a*, come back by *b*, and leave again by *c*. So N has 1 arrival and 2 departures, and 3 bridges touch it. That row is filled in for you. (If you could not find a route in Question 1(a), use this one.)

land	arrivals	departures	arrivals + departures	bridges
N	1	2	3	3
W				
S				
E				

Are the last two columns equal on every row? ☐ yes ☐ no

Question 3: Read your table and tick one box in each line.

(a) Two pieces of land have arrivals $\neq$ departures. They are _____ and _____.

(b) In your route, those two pieces of land are

☐ the first one and the last one ☐ two in the middle ☐ nothing special

(c) A piece of land that you only pass *through* has an

☐ even ☐ odd number of bridges.

(d) A route has one start and one finish. So a route crossing every bridge exactly once can exist only if the number of pieces of land with an **odd** number of bridges is

☐ 0 or 2 ☐ exactly 1 ☐ exactly 3 ☐ any number

(e) Count again on Map B, the eight-bridge map. It has _____ pieces of land with an odd number of bridges. Finish this sentence:

No route can exist on Map B because _____

Question 4: Königsberg, 1736. A real city with four pieces of land A, B, C, D and seven bridges *a* to *g*. Two pairs of bridges join the same two pieces of land; both bridges of a pair must be crossed.

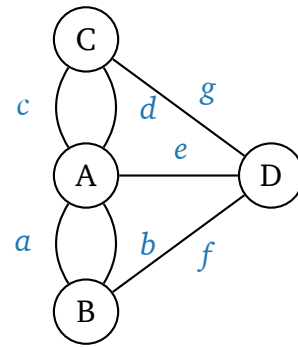
Count the bridges at each one:

A _____ B _____ C _____ D _____

Use your rule from 3(d). Can the walk be done?

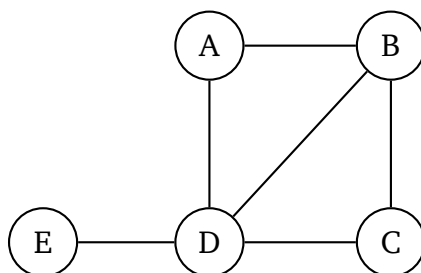
☐ yes ☐ no

In 1944 two bridges were destroyed and the walk became possible. Destroying just **one** bridge is already enough. List every single bridge that would work: _____



Part 2 — Three kinds of route

Forget bridges. Here is a different network: five bus stops, and the lines are direct bus services between them.



Question 5(a): For each route below, look at what it repeats. The first row is filled in for you.

route	repeats a line?	repeats a stop?	name
A B D B C	yes (B–D twice)	yes (B)	
A B D C B			
A B C D E			

Question 5(b): Here are the names. Write the right one in the last column of the table above.

A route that repeats **nothing** is a **path**.
 A route that may repeat stops but **never repeats a line** is a **trail**.
 A route that repeats **both** is just a **walk**.

Question 5(c): Write a route from A to E that repeats a stop but never repeats a line. (That is: a trail which is not a path.)

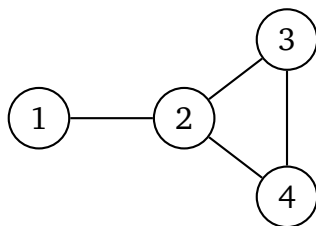
A _____ E

Question 5(d): Go back to your seven-bridge route in Question 1(a). Using these three names, it is a ☐ walk ☐ trail ☐ path.

Question 5(e): That route can never be a path. Finish the count that proves it: a path with 7 lines visits _____ different pieces of land, but the campus map only has _____.

Part 3 — Take these predictions to the laptop

A computer cannot look at a picture. Fill this page in **first**, on paper, then open the lab with your partner and find out if you were right. Here is a new network, with four nodes.



Question 6(a): Fill in the grid. Write **1** in row i , column j if a line joins node i and node j . Write **0** if it does not. Row 1 is done: node 1 is joined to node 2 only.

$A =$	1	2	3	4
1	0	1	0	0
2				
3				
4				

Question 6(b): Add up each row of your grid, and compare each total with the number of lines meeting that node in the picture.

row totals: _____

What do you find?

Question 6(c): A **2-step route** from node i to node j means: one line from i to somewhere, then one line from there to j . Count them on the picture, by hand.

2-step routes from node 1 to node 3: how many? _____ list them: $1 \rightarrow _____ \rightarrow 3$

2-step routes from node 1 to node 2: how many? _____ why? _____

2-step routes from node 2 back to node 2: how many? _____

Question 6(d): In the lab you will multiply your grid by itself, giving a new grid called A^2 . Write down, now, what you expect to find in it.

I expect row 1, column 3 of A^2 to be _____ row 1, column 2 to be _____

and the diagonal (row 1 col 1, row 2 col 2, ...) to be _____

Now open the lab, in pairs. One laptop between two people. The person who is *not* typing reads the question out loud and says what they expect before the other one runs the cell.

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Bring this sheet with you. The lab asks for the numbers you just wrote down.